Diagnosis on Winding Failure Through Impulsive Sound of Power Transformer

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***Abstract*—The transformer windings will flow a large impact current in the event of an external short circuit, causing strong vibration, resulting in winding looseness and deformation. Therefore, it is of great significance to carry out the research on fault diagnosis under short-circuit impact. The sound signal can reflect the overall mechanical condition of the equipment. According to the characteristics of the impulsive sound signal in time-frequency domain, two fault parameters like sound entropy (*SE*) and odd-even ratio (*OER*) were proposed. It is found that the *SE* and *OER* increase when the winding loosens, while they decrease when the deformation occurs. The deviation of parameters can reflect the degree of mechanical failure. According to the proposed fault parameters, a diagnosis on winding failure through the impulsive sound of the transformer is proposed.**

Keywords—power transformer; winding failure; impulsive sound; diagnosis

I. INTRODUCTION

As the main equipment for power transmission and distribution, transformers play a decisive role in power systems. According to the statistics, external short-circuits is one of the main factors that cause transformer failure[1][2]. In the course of long-term operation, there is an easy-to-short-circuit area in the power grid and the system is reclosed many times, so that the transformers will suffer repeatedly short-circuits, bringing irreversible cumulative looseness and deformation of the windings. Then the short-circuit withstand capability of the transformer is significantly reduced and the insulation ageing of the windings is accelerated simultaneously. Many transformers are facing these problems with high failure rates today, actions must naturally be taken.

At present, there are many detection methods for transformer winding deformation, among which there are low voltage impulse method (LVI)[3], short-circuit reactance measurement (SCR), frequency response analysis technique (FRA)[4], and sweep frequency impedance method (SFI)[5]. However, the above methods cannot perform live detection or on-line detection so their applications have been limited. The standard IEC 60076-5:2006 has been accepted and complied with by manufacturers and users. However, it is only judged from the deviaton of short-circuit reactance and has lower sensitivity to the faults. When an external short circuit occurs, a large impact current flows through wingdings causing strong vibration, which resulting in winding looseness and deformation. The vibration signal is closely related to the internal mechanical state of the transformer. Therefore, vibration signal analysis has received extensive attention due to its ease of on-line and live detection[6]. However, because of the complex structure of the transformer and the differences in the oil-tanks, it is difficult to select suitable measuring points. The acoustical signal which can reflect the overall mechanical condition of the equipment was utilized to conquer the shortcoming of conventional single-point vibration. The radiated sound signal also changes as the mechanical state of the winding changes.

In this paper, a 110kV power transformer was subjected to high-to-medium short circuit tests. According to the characteristics of the sound signal, a fault diagnosis method for winding failure based on short-circuit impulsive sound is proposed. The short-circuit reactance measurements and the final results of overhaul validated the accuracy of this new method. It provides a new idea for transformer mechanical fault diagnosis.

II. THE EXPERIMENTAL APPARATUSES AND PROGRAM

A. The Experimental Apparatuses

Short-circuit tests were carried on a 110 kV power transformer, as shown in Fig. 1, until the winding reactance changed more than 2%, of which phase A 33 times, phase B 8 times, phase C 16 times.



Fig. 1. Transformer and microphones

The measurement system is composed of microphones, a data acquisition card, and a laptop. It can achieve the trigger

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acquisition of burst signals. Two microphones were used to record the impulsive sounds on both sides of bushing outlet so that we can determine which side the fault occurred. The microphones were one meter away from transformer and located in the middle and one quarter of the oil-tank height as shown in Fig. 2.

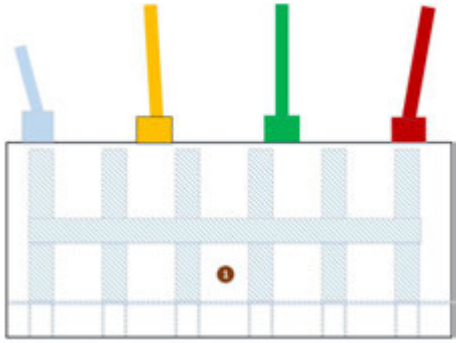


Fig. 2. The position of microphones

B. The Mechanism and Experimental program

The sound and vibration of power transformer in normal operation mainly come from the magnetostrictive of the core and the forced vibration of the windings in the leakage magnetic field. The fundamental frequency of the core vibration is twice as the excitation voltage frequency, and the acceleration is proportional to the square of the voltage; the fundamental frequency of the winding vibration is twice as the excitation current frequency, and the acceleration is proportional to the square of the current. When short-circuit fault occurs, the current flowing through winding can be up to nearly 30 times normal current, and the analytical expression is as in (1).

$$i_{\mathrm{k}} = -I_{\mathrm{m}}\left[\cos(\omega t + \alpha) - \cos\alpha \cdot e^{-\frac{R_{\mathrm{k}}}{L_{\mathrm{k}}}t}\right] \tag{1}$$

Where I_{m} is the maximum value of steady-state short-circuit current and $I_{\mathrm{m}} = U_{\mathrm{m}}/ z_{\mathrm{k}}$. Where U_{m} is the maximum system voltage and α is the phase of voltage. z_{k}, R_{k}, L_{k} are respectively short-circuit impedance, resistance and reactance.

According to Lorentz force law, the relationship between the electrodynamic force acting on the winding per unit length and the current is as in (2).

$$F(t) = kI_{\mathrm{m}}^2[\tfrac{1}{2} + \tfrac{1}{2}\cos 2(\omega t + \alpha) + \cos^2\alpha \cdot e^{-2\frac{R_{\mathrm{k}}}{L_{\mathrm{k}}}t}$$

$$-2\cos(\omega t + \alpha)\cdot\cos\alpha \cdot e^{-\frac{R_{\mathrm{k}}}{L_{\mathrm{k}}}t}] \tag{2}$$

Where k is electromagnetic force coefficient. The most serious situation occurs when α=0. And the instantaneous value of the short-circuit current is reached maximum when t =0. Sound pressure, vibration velocity, vibration acceleration, force, and current have the following relationship, as in (3). Therefore, when short circuit occurs, the sound pressure increases significantly, and the background noise has little effect on the impulsive sound.

$$p \propto v \propto a \propto f \propto i_{\mathrm{k}}^2 \tag{3}$$

Short-circuit tests were carried out on a transformer according to the standard IEC 60076-5：2006（Power transformers-Part 5:Ability to withstand short circuit）. The test numbers and the currents before short-circuit fault are given in the Table I. The duration of each test is 0.25 seconds. Unfortunately, the numbers in parentheses in the table refer to uncollected data.

TABLE I. CURRENTS AND TEST NUMBERS

PHASE A

Rated current ratio	70%	75%	80%	85%	90%
No.	1~3	4~6	7~9	10~12	13~15
Rated current ratio	95%	100%	105%		95%
No.	16~18(17)[a]	19~21	22~24(22)[a]		25~33

PHASE B

Rated current ratio	70%	75%	80%	85%
No.	1~3	4	5(5)[a]	6~8(6)[a]

PHASE C

Rated current ratio	70%	75%	80%	85%	90%	95%
No.	1~3	4~6	7~8	9~11	12~14	15~16

[a] Uncollected data.

III. ANALYSIS OF IMPULSIVE SOUND

A. Parameters of Impulsive Sound

When a short circuit occurs, the windings were subjected to large electromagnetic forces. After 0.25 s, the short-circuit current disappeared, and the forced vibration became free vibration. The sound pressure also gradually decayed with vibration. The signal of first short-circuit test on phase A is shown in Fig. 3.

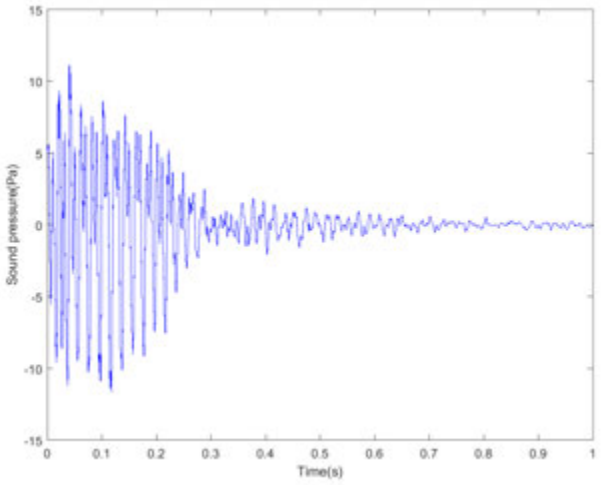


Fig. 3. The signal of impulsive sound

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Time-frequency analysis of the signal using complex wavelet transformer algorithm shows that frequencies of impulsive sound are mainly concentrated at 50 Hz and its harmonics, as shown in Fig. 4. When the machenical conditions of the windings changed, the time-frequency figure also changed obviously as shown in Fig. 5.

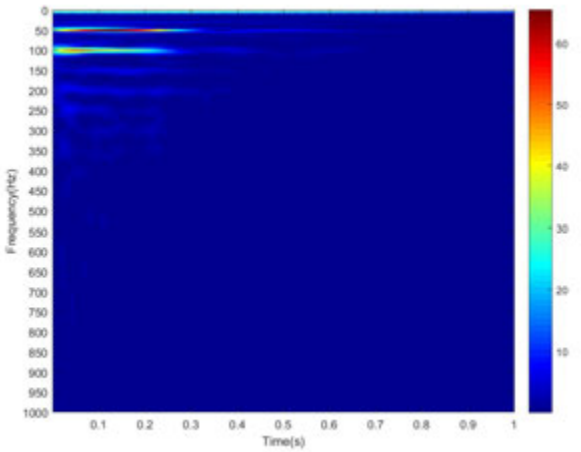


Fig. 4. Time-frequency analysis of signal under normal condition

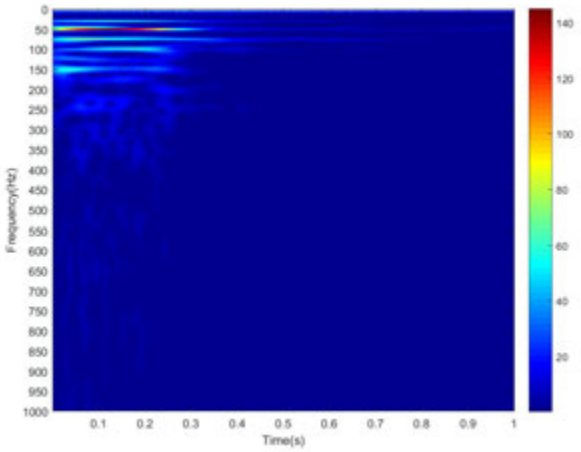


Fig. 5. Time-frequency analysis of signal after 23 short-circuit tests

Traditional signal analysis methods generally use Fourier analysis, which can not reflect the non-stationary, short duration, time domain and frequency domain localization characteristics of the signal. Wavelet analysis is a time-frequency method. Since it only decomposes the low-frequency band but considers less about the high-frequency band. The wavelet packet decomposition conquers this shortcoming and is a more sophisticated decomposition method. According to the characteristics of the sound signal in time-frequency domain, two fault parameters like sound entropy(*SE*) and odd-even ratio(*OER*) were proposed. Decompose the impulsive sound signal using 3-layer wavelet packet decomposition and abtain 8 frequency bands of decomposed signals. The wavelet basis is db5. The sampling rate is 1024 S/s, sampled for 1 second. Indices of terminal nodes is arranged in frequency order of 7,8,10,9,13,14,12,11. Calculate the energy of each band as E_7, E_8, E_{10}, E_9, E_{13}, E_{14}, E_{12}, E_{11}. Then the sound entropy(*SE*) and odd-even ratio(*OER*) were defined as follows.

$$SE = -\sum\left(\frac{E_{\mathrm{i}}}{E_{\mathrm{all}}}\log_2\frac{E_{\mathrm{i}}}{E_{\mathrm{all}}}\right) \tag{4}$$

$$\mathrm{OER} = \sum\frac{E_{\mathrm{j}}}{E_{\mathrm{all}}}\Big/\sum\frac{E_{\mathrm{k}}}{E_{\mathrm{all}}} \tag{5}$$

Where i=7, 8, …, 14, j=7, 10, 13, 12, k= 8, 9, 14, 11, and E_{all} is summation of E_7, E_8, ..., E_{14}.

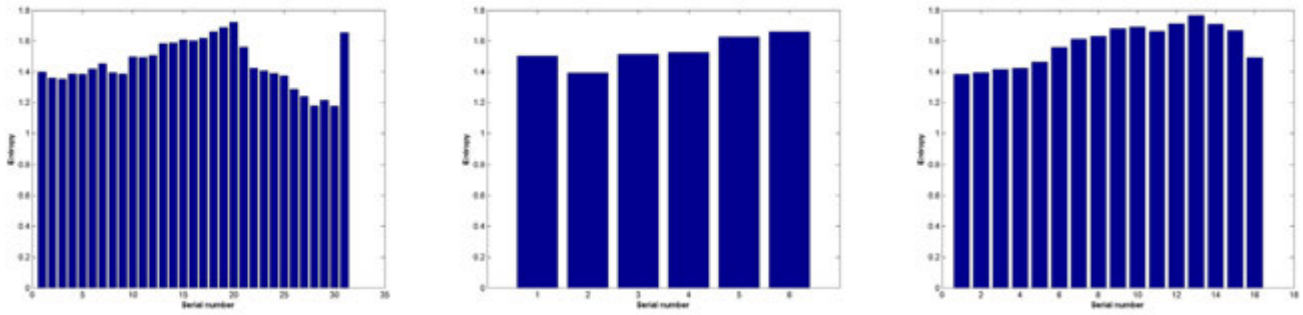
The *SE* indicates the degree of dispersion of energy. The more dispersed energy is, the larger *SE* is. The *OER* represents the proportion of odd frequency band energy. Both parameters are less affected by the current because they reflect the full frequency-domain information. When the winding is subjected to a large short-circuit current, force of 50 Hz and 100 Hz are generated, resulting in 50 Hz and 100 Hz vibration. In addition, due to the non-linearity of the spacer, higher harmonics are also generated. Then the 50 Hz vibration of conductor in the 50Hz nonuniform leakage magnetic field will couple out the electromechanical forces of 50Hz and its harmonics. As the winding becomes looser and looser, this coupling phenomenon becomes more and more obvious. In this case, the *SE* and *OER* become larger as well.

B. The Results of Short-circuit Tests

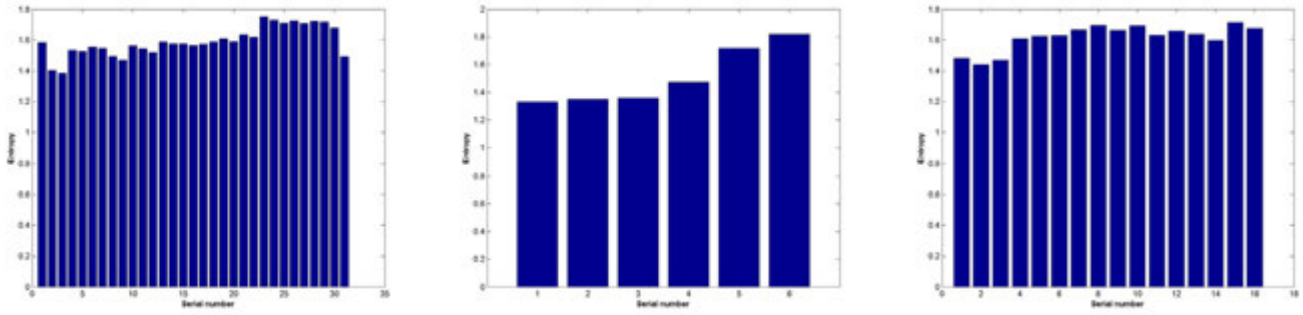
When the winding is loose, it becomes more deformable. On the other hand, when deformation occurs, the winding may become tighter in the early period or get looser in the later period. According to the analysis, an increase in *SE* or *OER* indicates wingding looseness, and a decrease indicates winding deformation. In addition, looseness and deformation may occur at the same time. As shown in Fig. 6 and Fig. 7, two parameters had a tendency to increase in the initial period which indicates that there was a gradual accumulation of looseness. The change extents of parameters can reflect the degree of changes in the mechanical state of the windings. In addition, the signal on different sides of transformer can evaluate which side of winding the faults occur on.

Take the impulsive sound signal of phase A at front side as an example. From the 1st to 22nd tests (No.1-No.20), the winding gradually loosened but the degree of deformation was not serious. Then the *SE* values of 23rd to 32nd (No.21-No.30) tests decreased and the *OER* increased, so that the winding were loose and deformed at the same time. After the 33rd (No.31) test, the *SE* increased 36.4% and the *OER* decreased 81.6%. The winding completely damaged and insulation fault may occur. The reactance diviation reached 21.85%.

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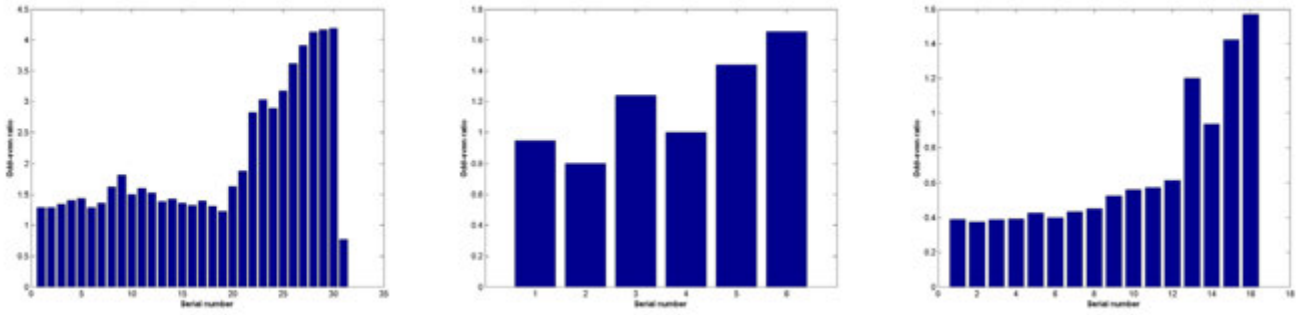


(a) Phase A at the front side (b) Phase B on the front side (c) Phase C on the front side

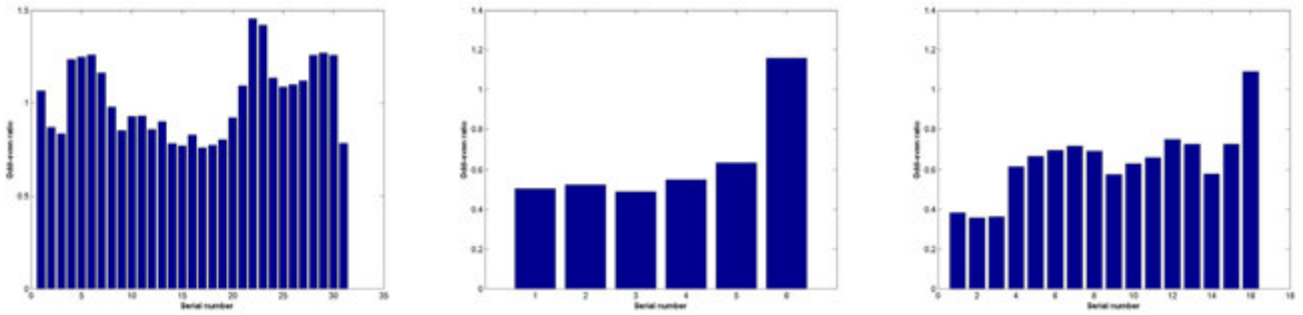


(d) Phase A on the rear side (e) Phase B on the rear side (f) Phase C on the rear side

Fig. 6. *SE* of every tests on both side



(a) Phase A on the front side (b) Phase B on the front side (c) Phase C on the front side



(d) Phase A on the rear side (e) Phase B on the rear side (f) Phase C on the rear side

Fig. 7. *OER* of every tests on both side

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C. Verification of Results

After each short-circuit test, phase reactance was measured. As shown in Table II, the reactance changed so weak that it cannot detect the minor fault unless the windings completely damaged. The experiments were performed until the deviation of the reactance exceeded 2%. Compared with *SCR*, the *SE* and *OER* are more sensitive and accurate to the mechanical conditions of the windings. They can assess what specific kind of fault the winding is, like looseness, deformation, or both.

TABLE II. REACTANCE AFTER SHORT CIRCUIT TEST

PHASE A

No.	0th	1st	9th	17th	25th	33rd
Reactance/Ω	23.77	23.87	23.86	23.89	24.04	18.58
Deviation/%	0	0.42	0.40	0.49	1.02	-21.85

PHASE B

No.	0th	1st	3rd	4th	7th	8th
Reactance/Ω	26.04	26.18	26.18	26.18	26.19	26.59
Deviation/%	0	0.54	0.54	0.54	0.58	2.12

PHASE C

No.	0th	1st	5th	9th	13th	16th
Reactance/Ω	16.69	16.81	16.81	16.75	16.79	18.18
Deviation/%	0	0.72	0.72	0.34	0.55	8.88

The final results of overhaul are shown in Fig. 8, Fig. 9 and Fig. 10. High-voltage winding of phase A was axially destabilized and concaved. Winding in the vicinity of severe deformation had different degrees of lodging. Medium-voltage winding of phase A was in good mechanical condition.

High-voltage winding of phase B was in good mechanical condition while medium-voltage winding were loose and concaved.

High-voltage winding of phase C was in good mechanical condition while medium-voltage winding was deformed severely.



Fig. 8. High voltage winding of phase A



Fig. 9. Medium voltage winding of phase B



Fig. 10. Medium voltage winding of phase C

IV. CONCLUSION

This paper proposed a new fault diagnosis method for winding failure based on short-circuit impulsive sound. Some results of this study can be summarized as follows. According to the characteristics of the impulsive sound in time-frequency domain, two fault parameters like sound entropy (*SE*) and odd-even ratio (*OER*) were proposed. The *SE* indicates the degree of dispersion of energy. The *OER* represents the proportion of odd frequency band energy. An increase in *SE* or *OER* indicates wingding looseness, and a decrease indicates winding deformation. Winding faults caused by short-circuit impact of transformer have cumulative effects. And this new method is more sensitive and accurate to the mechanical conditions of the windings than short-circuit reactance measurement. Further research is needed so that more accurate judgment basis can be put forward.

REFERENCES

[1] WG 12.19. "The short-circuit performance of power transformers," CIGRE Technical Brochure, Paris:Cigre,June 2002, pp.5-7.

[2] Wang, M., A. J. Vandermaar, and K. D. Srivastava. "Review of condition assessment of power transformers in service." IEEE Electrical Insulation Magazine 18.6(2003):12-25.

[3] Drobyshevski, A. A. "Assessment of transformer winding mechanical condition by low-voltage impulse method." Power Tech Conference Proceedings, 2003 IEEE Bologna IEEE, 2003:6 pp. Vol.2.

[4] Bagheri, Mehdi, et al. "Frequency response analysis and short-circuit impedance measurement in detection of winding deformation within power transformers." IEEE Electrical Insulation Magazine 29.3(2013):33-40.

[5] Liu, Yong, et al. "A study of the sweep frequency impedance method and its application in the detection of internal winding short circuit faults in power transformers." IEEE Transactions on Dielectrics & Electrical Insulation 22.4(2015):2046-2056.

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[6] Zhang, Fan, et al. "Frequency Response Function of Short Circuit Vibration for Power Transformer." Journal of Xian Jiaotong University (2017).

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